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Subject: Programming with Java	**Semester: III**
ASSIGNMENT No : 4	**Year: AY 2026-27**
TITLE : Wrapper Classes in Java	

Problem Statement

Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Learning Objectives

To perform conversion between primitive data types and wrapper objects using Java wrapper classes.

Program:

WrapperStudent.java

```
// Assignment 4 - Exercise 1: Wrapper class conversion for student marks

import java.util.*;

public class WrapperStudent

{

   public static void main(String args[])

   {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Subject 1 Marks: ");

        String m1 = sc.next();

        System.out.print("Enter Subject 2 Marks: ");

        String m2 = sc.next();
```

```
        Integer marks1 = Integer.valueOf(m1); // Converting String to
Integer object

        Integer marks2 = Integer.valueOf(m2); // Converting String to
Integer object



         int total = marks1 + marks2; // Auto-unboxing and arithmetic
addition

       System.out.println("Total Marks: " + total); // Displaying total

       sc.close(); // Closing scanner

   }

}
```

EmpPayrollSystem.java

```
// Assignment 4 - Exercise 2: Employee Payroll System using wrapper
validation
import java.util.Scanner;
public class EmpPayrollSystem
{
   public static void main(String args[])
   {
       Scanner sc=new Scanner(System.in);

       System.out.println("Enter The Following Details:");
       System.out.print("Employee ID: ");
       int empid=sc.nextInt();
       System.out.print("Basic Salary: ");
       double basic=sc.nextDouble();
       System.out.print("Bonus Amount: ");
       double bonus=sc.nextDouble();
       System.out.println("");

       double deductions=0.0;
        System.out.println("Enter Whether Using New Tax Regime or Old Tax
Regime:");
       System.out.println("'1' For New Tax Regime");
       System.out.println("'2' For Old Tax Regime");
       int choice=sc.nextInt();
       sc.close();
```

```
        System.out.println("");

        if(choice == 1)
        {
            deductions = basic * 0.05; // 5% flat tax deduction
                    System.out.println("New Tax Regime Applied (5% Tax
Deduction).");
        }
        else if(choice == 2)
        {
            deductions = basic * 0.10; // 10% flat tax deduction
                   System.out.println("Old Tax Regime Applied (10% Tax
Deduction).");
        }
        else
        {
                  System.out.println("Invalid Input. Select One Tax Slab
Please.");
        }
        // Primitive Type To Wrapper Object
        Integer empidobj=empid;
        Double basicobj=basic;
        Double bonusobj=bonus;

        if(empidobj <= 0)
        {
             System.out.println("Invalid. Employee ID must be greater than
0.");
        }
        else if(basicobj <= 0)
        {
                System.out.println("Invalid. Basic Salary must be greater
than 0.");
        }
        else if(bonusobj <= 0)
        {
                System.out.println("Invalid. Bonus Amount must be greater
than 0.");
        }
        else
        {
            double net = basicobj + bonusobj + deductions;
            System.out.println("\n- Employee Payroll Summary -");
            System.out.println("Employee ID: " +empidobj);
```

```
            System.out.println("Basic Salary: ₹" +basicobj);
            System.out.println("Bonus: ₹" +bonusobj);
            System.out.println("Tax Deductions: -₹" +deductions);
            System.out.println("Net Salary: ₹" +net);
        }
    }
}
```

Output:

WrapperStudent.java

```
PROBLEMS 4  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS                                                    Code

cd "/Users/thegharkalaptop/Documents/Programming (The RS)/" && javac WrapperStudent.java && java WrapperStudent
thegharkalaptop@The-Ghar-Ka-Laptop Programming (The RS) % cd "/Users/thegharkalaptop/Documents/Programming (The RS)/" && javac WrapperStudent.java && java Wrapper
Student
Enter Subject 1 Marks: 100
Enter Subject 2 Marks: 100
Total Marks: 200
thegharkalaptop@The-Ghar-Ka-Laptop Programming (The RS) %
```

EmpPayrollSystem.java

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS                                                    Code

cd "/Users/thegharkalaptop/Documents/Programming (The RS)/" && javac EmpPayrollSystem.java && java EmpPayrollSystem
thegharkalaptop@The-Ghar-Ka-Laptop Programming (The RS) % cd "/Users/thegharkalaptop/Documents/Programming (The RS)/" && javac EmpPayrollSystem.java && java EmpPa
yrollSystem
Enter The Following Details:
Employee ID: 101
Basic Salary: 10000
Bonus Amount: 5000

Enter Whether Using New Tax Regime or Old Tax Regime:
'1' For New Tax Regime
'2' For Old Tax Regime
1

New Tax Regime Applied (5% Tax Deduction).

- Employee Payroll Summary -
Employee ID: 101
Basic Salary: ₹10000.0
Bonus: ₹5000.0
Tax Deductions: -₹500.0
Net Salary: ₹15500.0
thegharkalaptop@The-Ghar-Ka-Laptop Programming (The RS) % cd "/Users/thegharkalaptop/Documents/Programming (The RS)/" && javac EmpPayrollSystem.java && java EmpPa
yrollSystem
Enter The Following Details:
Employee ID: 102
Basic Salary: 10000
Bonus Amount: 5000

Enter Whether Using New Tax Regime or Old Tax Regime:
'1' For New Tax Regime
'2' For Old Tax Regime
2

Old Tax Regime Applied (10% Tax Deduction).

- Employee Payroll Summary -
Employee ID: 102
Basic Salary: ₹10000.0
Bonus: ₹5000.0
Tax Deductions: -₹1000.0
Net Salary: ₹16000.0
thegharkalaptop@The-Ghar-Ka-Laptop Programming (The RS) %
```

Conclusion:

In this assignment, the concepts of data conversion and object encapsulation using wrapper classes were successfully implemented and verified. Wrapper classes provided reliable mechanisms for parsing string inputs into numeric formats, and autoboxing allowed seamless arithmetic operations between primitive data types and their corresponding object representations.